Week 11 – The Role of AI in Research: Generative AI (Lean Notes)

No transcript was available for this week, so these notes are built from the Week 11 assignment and its answer key only. They cover everything the assignment tests; lecture detail beyond it may be missing. ⭐ = tested in the assignment (every card is).

1. Concept cards

1. Machine learning is much older than generative AI ⭐

- **Arthur Samuel's checkers program** (1950s) improved by **playing many games** and adjusting from the outcomes. It learned **from experience**, is an early form of **reinforcement learning**, and was **not** given every possible strategy in advance.
- So AI-based learning did **not** begin with generative AI.

2. Classical AI vs generative AI ⭐

- **Classical AI** methods are still effective, often better and cheaper, for **narrow, well-defined, domain-specific** problems. They did not become irrelevant after LLMs.
- **Choose the approach to fit the problem**; generative AI is not "the only" or "always" right choice.

3. How an LLM writes: next-token prediction ⭐

- A **token** is a word or word-piece. An **LLM** (large language model) gives every possible next token a **probability that depends on the context** (the text so far), **picks one, appends it, and repeats**, building a long response.
- A different context gives a **different probability distribution**. The model can **sample**, so the highest-probability token is **not forced**; this is why answers vary.

4. Three size words ⭐

Term	Meaning
Token size (vocabulary)	the words or sub-words the model can generate
Context size	how much **previous text** the model can take into account at once
Parameter count	the learned numbers that **hold the learned probability patterns**

- Trap: the number of tokens in a prompt is the **context size**, not the parameter count.

5. Vector representations ⭐

- Storing a probability for every possible context is impossible, so a huge probability table is **compressed into a smaller vector space. Similar contexts land near each other.**
- Cost: because information is shared, one context can **influence another**, which causes **variability and errors**. Contexts are **not** kept as fully independent points.

6. Alignment ⭐

- **Alignment** adjusts a trained model so that its output probabilities better match **desired outcomes**: **factuality, bias reduction, local context**. A common method is **reinforcement learning from human preferences (feedback)**.

- It **cannot guarantee** that every output is factually correct.

7. Retrieval-Augmented Generation (RAG) ⭐

- **RAG: retrieve** relevant information from a repository and **add it to the original question** before generating the answer. It supplies **extra context** and makes the response more focused.
- It **does not retrain** the model and **does not increase the number of parameters**; only the input changes.
- **2011 ICC World Cup example:** "who won the World Cup" is ambiguous; adding the year and event gives context that **moves the question to a better place in the vector space**, so the intended answer becomes more probable.

8. Hallucination ⭐

- **Hallucination** = a **plausible-sounding answer that does not match the ground truth**. It is more likely when the model has **thin information or context**.
- **Larger context can reduce it, but it cannot necessarily be eliminated.** The model has **no intention to deceive**.

9. Misinformation from sources vs hallucination ⭐

- If a false or misleading article was in the data the model learned from, the model can **repeat misinformation**. This is **not necessarily hallucination**, which is inventing an unsupported fact.
- Either way, **important AI output needs independent verification**.

2. Top traps (most likely first)

1. **Absolute words** ("only", "always", "must always", "guarantees", "necessarily", "completely independent") make an option wrong.
2. The **highest-probability token is not always generated**; the model can sample.
3. **Context size is not parameter count.**
4. **RAG changes the input, not the model** (no retraining, no new parameters).
5. **Alignment reduces errors but never guarantees correctness.**
6. **Hallucination is not lying**, and a model repeating a false source is not always hallucinating.
7. **Samuel's checkers (1950s) is early reinforcement learning**, learned by playing, not pre-programmed.
8. **Classical AI is not obsolete**; pick the tool by the problem.
9. The portal cuts off Q8 option D ("…producing the intended"); it presumably ends "answer" and is correct.

3. Assignment questions with solutions

Q1. Arthur Samuel's checkers program illustrates an early form of learning by machines. Which statements are correct? (select all that apply) *(Q1)* - A. The machine was explicitly programmed with every possible checkers strategy B. The machine learned through experience by repeatedly playing the game C. The example is associated with an early instance of reinforcement learning D. The example demonstrates that AI-based learning is a phenomenon that began only with generative AI

Solution (card 1): A is false (it learned by playing), D is false (1950s). **Answer: B and C.**

Q2. Which statements correctly reflect the discussion of classical AI and generative AI? (select all that apply) *(Q2)* - A. Generative AI is the only appropriate approach for solving modern research problems B. Classical AI methods can still provide effective solutions for domain-specific problems C. Researchers should select an AI approach according to the problem rather than assuming that generative AI must always be used D. Classical AI became irrelevant after the emergence of LLMs

Solution (card 2): A and D are absolute. **Answer: B and C.**

Q3. Consider an LLM that predicts the next token from the context provided to it. Which statements are correct? (select all that apply) *(Q3)* - A. The probability of a token can depend on the preceding context B. A token with the highest probability must always be generated C. Different contexts can result in different probability distributions for the next token D. The model can repeatedly predict tokens to construct a longer response

Solution (card 3): B is wrong because the model can sample. **Answer: A, C and D.**

Q4. Which correctly distinguish token size, context size, and parameter count in an LLM? (select all that apply) *(Q4)* - A. Token size is related to the vocabulary of words or subwords that the model can generate B. Context size refers to the amount of previous token information considered by the model C. Parameter count refers to the number of tokens that can be stored in the prompt D. Parameter count is related to how the learned probability distributions are represented in the model

Solution (card 4): C describes context size. **Answer: A, B and D.**

Q5. Which statements correctly describe the role of vector representations? (select all that apply) *(Q5)* - A. A very large probability matrix can be represented through a comparatively smaller-dimensional vector space B. Different contexts may be projected to nearby locations in the vector space C. Such compression can contribute to variability and errors because information from different contexts may influence generation D. Every possible context is retained as a completely independent point with no overlap

Solution (card 5): D would mean no compression. **Answer: A, B and C.**

Q6. Which statements are consistent with the explanation of alignment? (select all that apply) *(Q6)* - A. Alignment can modify model behaviour so that generated probabilities better correspond to desired outcomes B. Alignment may be directed towards objectives such as factuality, bias reduction or local context C. Alignment guarantees that all outputs produced by an LLM are factually correct D. Reinforcement learning from human preferences is associated with the alignment process

Solution (card 6): C says “guarantees”. **Answer: A, B and D.**

Q7. A researcher asks a small language model an ambiguous question. Relevant information is retrieved from an internal repository and added to the original question before generating the answer. Which statements are correct? (select all that apply) *(Q7)* - A. The approach is consistent with Retrieval-Augmented Generation (RAG) B. The retrieved information provides additional context to the model C. The approach necessarily requires retraining the entire language model D. Additional context can make the model’s response more focused

Solution (card 7): only the input changes. **Answer: A, B and D.**

Q8. The 2011 ICC World Cup example explains the effect of RAG. Which correctly captures the idea? (select all that apply) *(Q8)* - A. Additional retrieved information can contextualize an otherwise ambiguous query B. The original question can be effectively shifted toward a more appropriate contextual representation C. RAG works primarily by increasing the number of parameters in the LLM D. A richer context can increase the probability of producing the intended (answer, cut off by the portal)

Solution (card 7): C is wrong since parameters do not change. **Answer: A, B and D.**

Q9. Which statements about hallucination are correct? (select all that apply) *(Q9)* - A. A model may produce a plausible answer even when the generated information does not match the ground truth B. Hallucination can occur when the model has insufficiently represented information or context C. Hallucination can be reduced through techniques such as providing larger context, but it cannot necessarily be eliminated completely D. Hallucination means that the model has deliberately decided to provide false information

Solution (card 8): D wrongly claims intent. **Answer: A, B and C.**

Q10. A model produces a misleading recommendation because a false or deliberately misleading article was present in the information it learned from. Which statements are appropriate? (select all that apply) *(Q10)* - A. The problem is necessarily an example of hallucination B. AI systems can potentially propagate misinformation present in their information sources C. A response may be misleading even when it is not simply an invented fact by the model D. Independent verification of important AI-generated information remains necessary

Solution (card 9): A says “necessarily”. **Answer: B, C and D.**